Building a Chatbot using LangGraph

Tuesday, March 24, 2026 6:01 PM

Recap

Another library

↳ LangGraph vs Langchain

Every app graph
↓
In comp with
Other application
↓ agents
↓
Control is high
Autonomy is high.

Graph
↓
Any ele in the graph cannot interact with any other element in graph with help of 'edges'
↓
Build very complex applications.

ADV

Ch✓
add✓T
add MV
Create✓

LCEL Chain
↓
Sequence runnables elem
↓
Branch parallelize

Chatbot
add tools
add memory
create an agent

An element in LCEL cannot interact with all the the ele in chain✓

Lang Graph → Graph
↓
Nodes, edges
↓ perform operations
↓ connects nodes. [Sequence] path

START ✓

Any Number of nodes, connected by edges with any complex interactions

END.

{'var' : 'Hello'} State → Python obj ⇒ Dictionary.

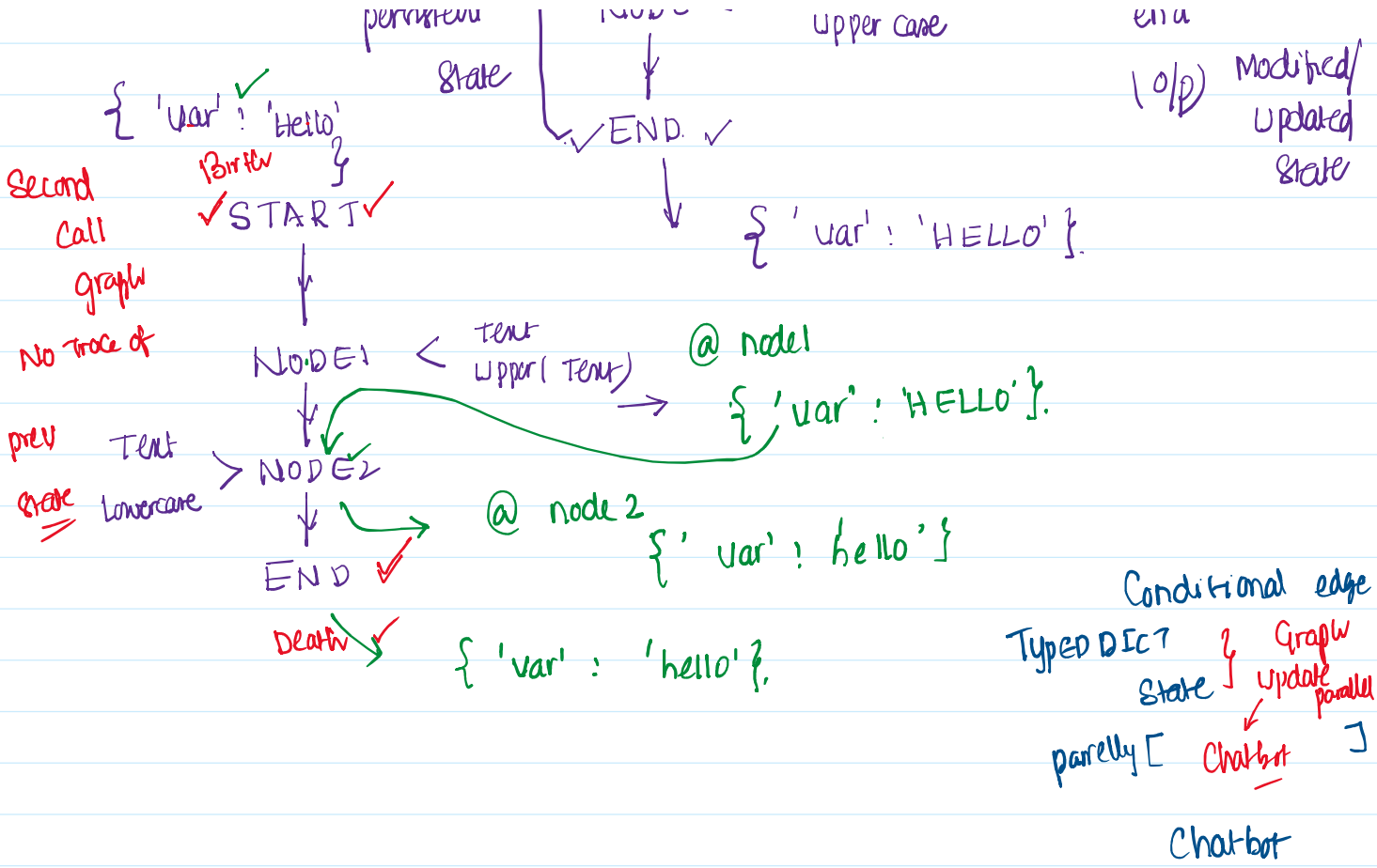
State persistent state

START ✓
↓
NODE ✓ ← Manage upper case
↓

State (I/P) Schema → persistent state

Start
↓
all nodes in LangGraph interact state
↓
Nodes
↓
end

(O/P) Modified/


persistent state
upper case
end
(o/p) Modified/ updated state
{ 'var' : 'Hello' }
Birth
Second call graph
START
END
{ 'var' : 'HELLO' }
No trace of prev state
NODE1
Text
upper(Text)
@ node1
{ 'var' : 'HELLO' }
Text
Lowercase
NODE2
@ node 2
{ 'var' : 'hello' }
END
Death
{ 'var' : 'hello' }
Conditional edge
TypeDICT
State
Graph update
parallel
parelly [Chatbot]
Chatbot